Zephaniah Roe

Chicago, Illinois | zroe@uchicago.edu | github.com/zroe1

Education

The University of Chicago — Chicago, IL
Bachelor of Science in Computer Science; GPA: 3.85/4.0 — *Expected June 2026*

Experience

University of Chicago Existential Risk Lab — Chicago, IL
AI Security Team Lead / AI Risk Student Lead — *August 2024 – Present*

- Led the effort to create the XLab AI Security Guide, a resource for learning about jailbreaks and adversarial robustness
- Coded the website for the guide and led the effort to design and code the xlab-security Python package
- Wrote prose and designed supplemental coding exercises for many sections of the website

University of Chicago Conceptualization Lab — Chicago IL
Student Researcher for Ari Holzman — *October 2024 – Present*

- Performed experiments with Tinker and Together to quantify how much LLMs transfer generalized personas learned in post-training
- Quantified the conditions under which LLMs learn generalized personas or rules from narrow datasets

Supervised Program for Alignment Research (SPAR) — Remote
Research Fellow — *January – September 2025*

- Performed activation patching and trained linear probes to study how diffusion policy robots make decisions
- Researched small autoencoders to study "superposition" in neural networks

JusticeText — Remote
Software Engineering Intern — *June – August 2024*

- Implemented React UI components and performed frontend optimizations to reduce API calls and latency
- Developed AWS DynamoDB transaction logic and Node.js backend APIs

University of Chicago Harris School of Public Policy — Chicago, IL
Research Assistant — *January – June 2024*

- Designed an automated pipeline scrape Afghan government websites and download PDFs for name extraction
- Employed Anthropic and OpenAI APIs to extract names from Afghan government documents to track employment trends
- Performed statistical analysis on metadata present in images, PDFs, and Excel files from Afghan ministries

Programming

Languages: Python, C, Rust, JavaScript, TypeScript

Technologies: PyTorch, Tinker, Together, OpenAI/Anthropic APIs, OpenRouter, Git, DynamoDB, Linux, LaTeX

Projects:

- What Washington Says About AGI: A blog post summarizing research on every congressperson's position on AI
- Blog post on GPT-OSS jailbreaks: Investigation of chain-of-thought jailbreaks on OpenAI open source models
- Reevaluating "Model Organisms of Emergent Misalignment": A replication of a paper on Emergent Misalignment
- Toy Models of Superposition Paper Replication: Replication of an Anthropic research paper
- Neural network from scratch: Neural network implemented with only the standard library in C

Activities

UChicago Institute of Politics TechTeam / Genesys Works — Project Co-Leader, January – October 2024

- Led a team of over 10 undergraduate and masters students to build software for the non-profit Genesys Works
- Oversaw the codebase used in three major US cities to pair hundreds of low-income students with internships

UChicago Alt Protein Project — Research Leader, September 2024 – January 2025

- Led the computational/ML project for UChicago's alternative protein research accelerator
- Advised a group of 5 undergraduates and developed a novel approach to protein classification

Other Activities: UChicago Effective Altruism, Boxing Club, Rationality Club
Interests: Latin prose and poetry, prediction markets, boxing, balloon twisting